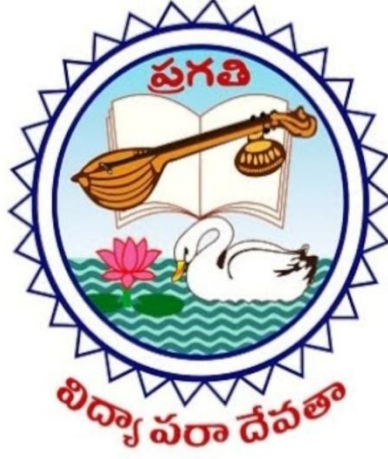




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## **FOOD BANK INVENTORY MANAGEMENT SYSTEM**

### **. Introduction**

The Food Bank Inventory Management System is a console-based application developed using C++. The main purpose of this system is to manage food items stored in a food bank efficiently. It helps users to add, view, search, update, and delete food item records easily.

This system reduces manual work and improves inventory management by maintaining food details such as Food ID, Food Name, Quantity, and Expiry Date in an organized manner.

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## **2. Objectives**

- To maintain food inventory records efficiently.
- To reduce manual errors in food management.
- To provide quick searching of food items.
- To update food quantities easily.
- To remove expired or unnecessary food records.
- To learn Object-Oriented Programming concepts using C++.

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## **3. System Features**

### **Main Features**

1. Add New Food Item
2. Display All Food Items
3. Search Food Item by ID
4. Update Quantity of Food Item
5. Delete Food Item
6. Menu Driven User Interface
7. Inventory Storage Using Arrays
8. Expiry Date Management
9. Easy Console-Based Operation

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## **4. Technologies and Concepts Used**

## Technologies Used

- Programming Language: **C++**
  - Platform: Console Application
  - Compiler: GCC / g++
  - IDE:
    - Visual Studio Code
    - Code::Blocks
    - Dev-C++
- 

## Concepts Used

- Classes and Objects
  - Arrays of Objects
  - Functions
  - Loops (for, do-while)
  - Conditional Statements (if, switch)
  - String Handling
  - Linear Search
  - Input and Output Streams
  - Object Assignment
- 

## 5. System Design

### Class Design

#### Class: Food Item

| DATA MEMBERS | DESCRIPTION      |
|--------------|------------------|
| ID           | STORES FOOD ID   |
| NAME         | STORES FOOD NAME |
| QUANTITY     | STORES QUANTITY  |

|             |                    |
|-------------|--------------------|
| EXPIRY DATE | STORES EXPIRY DATE |
|-------------|--------------------|

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## Member Functions

| FUNCTION        | PURPOSE              |
|-----------------|----------------------|
| ADD FOOD ()     | ADDS FOOD DETAILS    |
| DISPLAY FOOD () | DISPLAY FOOD DETAILS |

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## 6. Workflow

### Step-by-Step Working Process

1. Program starts.
  2. Main menu is displayed.
  3. User selects an option.
  4. Based on choice:
    - Add food item
    - View items
    - Search item
    - Update quantity
    - Delete item
  5. System performs selected operation.
  6. Menu repeats until user chooses Exit.
  7. Program terminates.
- 

## 7. Sample Output

### Main Menu

===== FOOD BANK INVENTORY MANAGEMENT =====

1. Add Food Item

2. View All Food Items

3. Search Food Item

4. Update Quantity

5. Delete Food Item

6. Exit

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### **Add Food Item**

Enter Food ID: 101

Enter Food Name: Rice Bags

Enter Quantity: 50

Enter Expiry Date: 20/12/2026

Food Item Added Successfully!

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### **View Food Items**

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Food ID : 101

Food Name : Rice Bags

Quantity : 50

Expiry Date : 20/12/2026

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### **Search Food Item**

Enter Food ID to Search: 101

Food Item Found Successfully!

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### **Update Quantity**

Enter Food ID to Update Quantity: 101

Current Quantity: 50

Enter New Quantity: 70

Quantity Updated Successfully!

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### **Delete Food Item**

Enter Food ID to Delete: 101

Food Item Deleted Successfully!

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**Exit**

Thank You for Using the System!